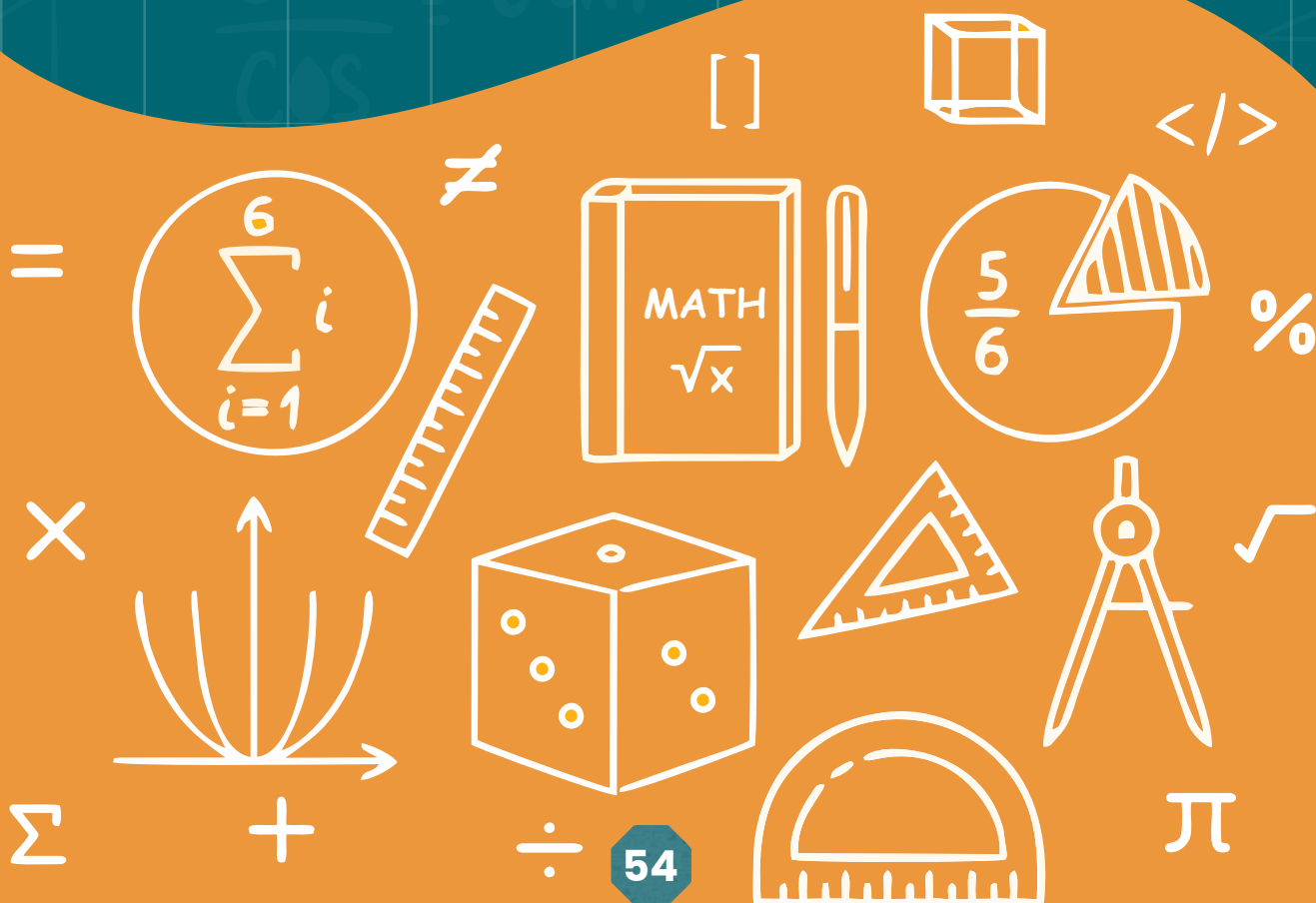


UNIT 6

DIRECT AND INVERSE VARIATION

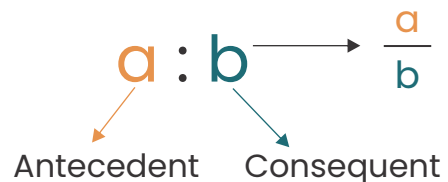


RATIO

A **ratio** is a way to compare two quantities of the same kind. It tells us **how much of one thing there is compared to another**.

Formula for a Ratio:

The formula for a ratio is written as $a : b$, which can also be written as $\frac{a}{b}$.



- **a** is the first term (called the **antecedent**).
- **b** is the second term (called the **consequent**).

A ratio is a comparison of two quantities of the same kind. It shows how much there is of one thing compared to another. The formula for a ratio is $a : b$, which is equivalent to $\frac{a}{b}$.

In this formula, a is the first term or antecedent, and b is the second term or consequent.

For example, if there are eight oranges and six lemons in a bowl of fruit, the ratio of oranges to lemons is $8 : 6$, which is equivalent to $4 : 3$.

Key Points to Remember:

- The ratio should exist between the quantities of the same kind
- While comparing two things, the units should be similar
- There should be significant order of terms
- The comparison of two ratios can be performed, if the ratios are equivalent like the fractions

Example 1:

Find the continued ratio $A : B : C$ if

- $A : B = 5 : 6$
- $B : C = 6 : 7$

Solution:

- Here $A : B = 5 : 6$ and $B : C = 6 : 7$.
 - Since the size of the common quantity B is the same in both ratios i.e. 6,
- Therefore $A : B : C = 5 : 6 : 7$.

Example 2:

Find the continued ratio $X : Y : Z$, if $X : Y = 6 : 7$ and $Y : Z = 8 : 9$

Solution: First Method

First of all, we find the product of the sizes of common quantity Y , which is 56. Now we make the size of Y the same (i.e., 56) in each ratio.

$$\begin{array}{llll}
 \text{Here} & X : Y = 6 : 7 & \text{and} & Y : Z = 8 : 9 \\
 \\
 \text{or} & \frac{X}{Y} = \frac{6}{7} & \text{or} & \frac{Y}{Z} = \frac{8}{9} \\
 \\
 & = \frac{6 \times 8}{7 \times 8} = \frac{48}{56} & & = \frac{8 \times 7}{9 \times 7} = \frac{56}{63}
 \end{array}$$

Hence $X : Y : Z = 48 : 56 : 63$

Second Method

$$\begin{array}{ccccc}
 X & : & Y & : & Z \\
 6 & & 7 & & \\
 & \swarrow & \updownarrow & \nwarrow & \\
 & & 8 & & 9 \\
 \hline
 (6 \times 8) & : & (7 \times 8) & : & (7 \times 9) \\
 48 & : & 56 & : & 63
 \end{array}$$

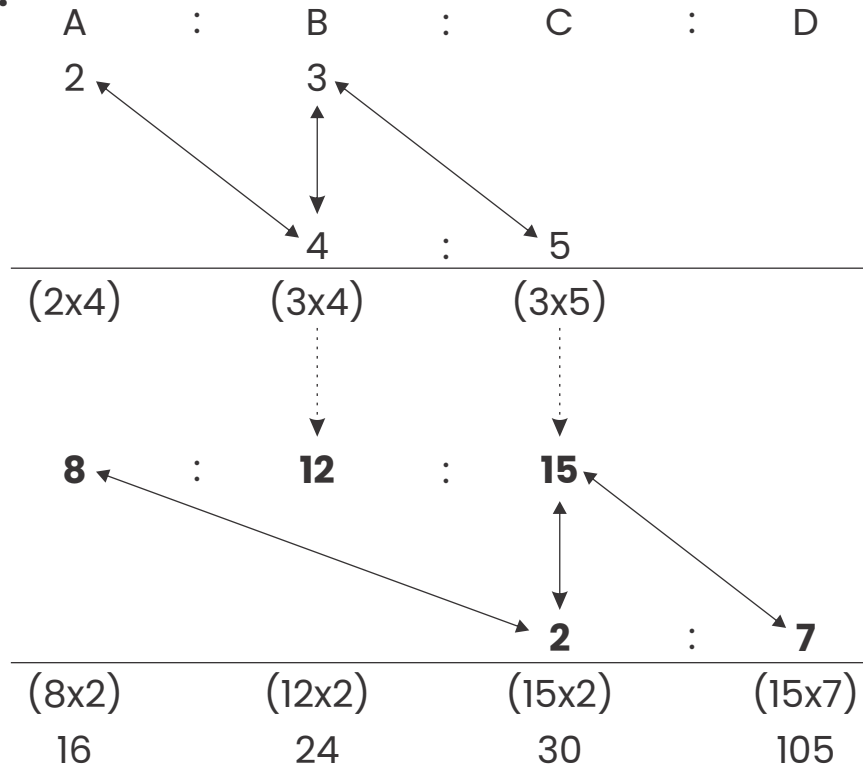
Hence $X : Y : Z = 48 : 56 : 63$

We can also find the continued ratio of four quantities as explained below:

Example 3:

Find $A : B : C : D$ if $A : B = 2 : 3$, $B : C = 4 : 5$ and $C : D = 2 : 7$

Solution:



Hence $A : B : C : D = 16 : 24 : 30 : 105$

PROPORTION

A proportion is a statement that shows two ratios are equal. It means that the relationship between one pair of numbers is the same as the relationship between another pair of numbers.

Symbols for Proportion:

Proportions are written using the symbols:

- $::$ (double colon)
- $=$ (equals sign)

Example:

If a train travels **100 km in 1 hour** and **500 km in 5 hours**, the proportion is:

$$\frac{100 \text{ km}}{1 \text{ hour}} = \frac{500 \text{ km}}{5 \text{ hours}}$$

This means the two ratios are equivalent.

Types of Proportion:

1. Direct Proportion

When one quantity **increases**, the other quantity also **increases**. Similarly, when one quantity **decreases**, the other quantity also **decreases**.

• **Representation:**

If **a** and **b** are two quantities, we write: $a \propto b$ (read as "a is directly proportional to b").

• **Example:**

If 1 book costs **100 rupees**, then 2 books will cost **200 rupees**.

- The cost increases as the number of books increases.

2. Inverse Proportion

When one quantity **increases**, the other quantity **decreases**. Similarly, when one quantity **decreases**, the other quantity **increases**.

• **Representation:**

If **a** and **b** are two quantities, we write: $a \propto \frac{1}{b}$ (read as "a is inversely proportional to b").

• **Example:**

If **5 workers** take **10 days** to complete a task, then **10 workers** will take **5 days** to complete the same task.

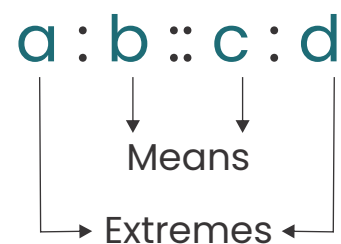
- As the number of workers increases, the time required decreases.

Proportion Formula

Now, let us assume that, in proportion, the two ratios are **a : b** & **c : d**. The two terms '**b**' and '**c**' are called "**means or mean term**", whereas the terms '**a**' and '**d**' are known as "**extremes or extreme terms**".

$$\frac{a}{b} = \frac{c}{d} \text{ or } a : b :: c : d$$

Product of extremes = Product of means
 $a \times d = b \times c$



Difference Between Ratio and Proportion

S. No.	Ratio	Proportion
1	The ratio is used to compare the size of two things with the same unit	The proportion is used to express the relation of two ratios
2	It is expressed using a colon (:), slash (/)	It is expressed using the double colon (::) or equal to the symbol (=)
3	It is an expression	It is an equation

Example 1: Solving a Proportion

Find x if $2 : 9 = x : 36$

Solution:

We have,

$$2 : 9 = x : 36$$

As, product of means = product of extremes

$$\text{So, } 9 \times x = 2 \times 36$$

$$\text{or, } \frac{9 \times x}{9} = \frac{2 \times 36}{9} = \frac{2 \times 36}{9} = 2 \times 4 = 8$$

Thus, $x = 8$

Proportion and Unitary Method

Example 1: If the cost of 10 similar books is 500 rupees, find the cost of 15 books.

Solution: Proportion Method

Let the cost of 15 books be x rupees.

Books	Cost (rupees)	
10	500	
15	x	(Direct Proportion)

$$10 : 15 = 500 : x$$

As product of means = product of extremes,

$$\text{So, } 10 \times x = 15 \times 500$$

$$\text{or, } \frac{10 \times x}{10} = \frac{15 \times 500}{10}$$

$$\text{or, } x = 750$$

So, the required cost is Rs 750.

Unitary Method

The cost of 10 books = 500 rupees

The cost of 1 book = $\frac{500 \text{ rupees}}{10}$ (Less, so divide)

= 50 rupees

The cost of such 15 books = 15×50

= 750 rupees

So the required cost of 15 books is Rs 750.

EXERCISE 6.1

1. Find A : B : C, if

(i) $A : B = 2 : 5$ and $B : C = 5 : 4$

(ii) $A : B = 3 : 7$ and $B : C = 8 : 9$

(iii) $A : B = 10 : 11$ and $B : C = 20 : 21$

2. Find A : B : C : D, if

(i) $A : B = 2 : 3$, $B : C = 4 : 5$ and $C : D = 7 : 10$

(ii) $A : B = 4 : 3$, $B : C = 5 : 7$ and $A : D = 7 : 9$

3. Find x, if

(i) $x : 2 = 5 : 6$

(ii) $3 : x = 9 : 7$

(iii) $4 : 5 = x : 7$

4. The ratio of weights of Shazia and Nazia is 5 : 9 and that of Nazia and Madhia is 8 : 11. Find the continued ratio of their weights.

5. Divide 2000 rupees among Saleem, Irfan and Imran in 2 : 3 : 5.

6. Solve the following problems by unitary method and proportion method:

1. The price of 15 chairs is 4500 rupees. Find the price of such 20 chairs.

2. Six identical pipes can fill a water tank in 3 hours. How many such pipes can fill the same tank in 2 hours?

3. If 5 persons can consume the food provisions in 8 days, how many persons can consume the same food provisions in 4 days?

4. The cost of 4 kilograms of mangoes is 300 rupees. Find the cost of 10 kilograms of mangoes.

5. A student reads 30 pages of a book in 4 days. How many pages of the same book will be read in 10 days?

TIME, WORK AND DISTANCE

Time:

When we solve the problems related to time and work, it is observed that:

(i) Time is directly proportional to work.

Because more work takes more time and less work takes less time.

(ii) Number of workers is inversely proportional to the time.

Because more workers take less time to complete the job. And less number of workers needs more time to complete the work.

Example 1: Ten persons can complete $\frac{1}{2}$ of a work in a certain time. How many persons can complete $\frac{3}{4}$ of the same work in the same time?

Solution: Let the required number of persons be x ,

Persons

10

 $\times$

Work

 $1\frac{1}{2}$ $3\frac{3}{4}$

(Direct Proportion)

We have,

$$10 : \times = \frac{1}{2} : \frac{3}{4}$$

As product of means = product of extremes,

$$\text{So, } \frac{1}{2} \times \times = 10 \times \frac{3}{4}$$

$$\text{or } \frac{\times}{2} = \frac{30}{4}$$

$$\text{or } 4x = 60$$

$$\text{or } x = 15$$

So, the required number of persons is 15.

Example 2: Six persons can complete a work in 7 days. How many persons can complete the same work in 6 days?

Solution: Let the required number of persons be x .

Persons

6

 $\times$

Days

7

6

(Inverse Proportion)

We have,

$$6 : \times = 6 : 7$$

As product of means = product of extremes,

$$\text{So, } 6 \times \times = 6 \times 7$$

$$\text{or } \frac{6 \times \times}{6} = \frac{6 \times 7}{6}$$

or $x = 7$

So, the required number of persons is 7.

Distance:

In daily life, we notice that distance increases or decreases proportionally with time.

Suppose a car covers 20 km in 10 minutes and 30 km in 15 minutes. If we find the relation or ratio of distance with time, we get:

$$\frac{20}{10} = \frac{30}{15} = 2$$

We see the ratio is the same, and this ratio of distance to time is called speed. In general, we write as:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Example 1: A car travels 15 km in 3 hours. Find its speed.

Solution: Here Distance = 15 km and Time = 3 hours.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{15}{3} = 5 \frac{\text{km}}{\text{hr}}$$

Example 2: The speed of a bus is 60 km/h. How much distance will it cover in 4 hours.

Solution: Here Speed = 60 $\frac{\text{km}}{\text{hr}}$, Time taken = 4 hours and Distance = ?

We know that:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{i.e. } 60 \frac{\text{km}}{\text{hr}} = \frac{\text{Distance}}{4 \text{ hr}}$$

$$4 \text{ hr} \times 60 \frac{\text{km}}{\text{hr}} = \text{Distance}$$

$$240 \text{ km} = \text{Distance}$$

So, the bus covers 240 km in 4 hours.

1 kilometre = 1000 metres and 1 hour = 3600 seconds

By using the above relation, we can find the relation between the units of speed i.e., between km/h and m/s.

The relation is: $1 \text{ km/h} = \frac{1000 \text{ m/s}}{3600}$

Example 1: Convert 72 km/h into m/s

Solution: $72 \text{ km/h} = 72 \times \frac{1000 \text{ m/s}}{3600} = \frac{720}{36} = \frac{720}{36} = 20 \text{ m/s}$

Example 2: Convert 5000 m/s into km/h

Solution: $5000 \text{ m/s} = 5000 \times \frac{3600 \text{ km/h}}{1000} = 18000 \text{ km/h}$

Example 3: A vehicle travels 120 km in 2 hours. How much distance will it cover in 6 hours?

Solution: Let the required distance be x kilometers.

Kilometres	Hours	
120	2	
x	6	(Direct Proportion)

We have, $120 : x = 2 : 6$

As product of means = product of extremes,

$$\begin{aligned} \text{So, } 2 \times x &= 6 \times 120 \\ x &= 360 \end{aligned}$$

So, the required distance is 360 km.

EXERCISE 6.2

- 1.** 20 workers can complete half of work in a certain time. How many workers are required to complete three-fourth of the work in the same time?
- 2.** If 6 labourers complete a work in 10 days. How many labourers are required to complete the same work in 15 days?
- 3.** If 5 workers can complete a work in 8 days. How many workers are required to complete the same work in 4 days?
- 4.** If three masons construct a wall in four days. In how many days, five such masons can construct the wall?
- 5.** If 5 men complete one third of a work. How much work will be completed by 7 men in the same time?
- 6.** A bus takes 3 hours to reach its destination if speed is 60 km/h. In how much time will it cover the same distance with the speed of 70 km/h?
- 7.** A vehicle takes 3 hours to complete the distance of 210 km.
 - (i)** What is its speed per hour?
 - (ii)** What is its speed in km/minute?
 - (iii)** What is its speed in m/s?
 - (iv)** What distance will it cover in 4.5 hours; with the speed of 70 km/h?
 - (v)** What time will it take to complete the distance of 350 km?
- 8.** Find the speed of a motorbike which covers a distance of 80 km in 2 hours.